

Nyx

A Data-Driven Astronomical Observation Platform Using NASA Space Data and Local Observations in Mauritius

16.01.2026

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Project Overview

Nyx is a data-driven astronomical observation platform that combines publicly available NASA space mission data with ground-based optical observations conducted in Mauritius.

The project focuses on designing a system for collecting, validating, visualising, and managing astronomical time-series data, while emphasising software architecture, data pipelines, and performance-aware system design.

The platform is intended as a proof-of-concept for how local scientific instrumentation can be integrated with large-scale open scientific datasets.

Problem Statement

NASA provides vast quantities of high-quality astronomical data through open APIs, but this data is typically consumed in isolation from local observational contexts.

In Mauritius, access to professional observatories is limited, and amateur or academic ground-based observations are rarely integrated into modern data platforms.

There is currently no unified system that:

- Integrates local telescope observations with space-based datasets
- Provides interactive, authenticated access to astronomical data
- Enables structured comparison and validation workflows

Project Objectives

The main objectives of this project are:

- To design a platform that integrates NASA open astronomical data with locally captured telescope observations
- To explore data validation through cross-source comparison
- To design a performant backend suitable for scientific workloads
- To create a modern, reactive frontend for data exploration
- To demonstrate how data science principles apply beyond traditional datasets

Data Sources

The platform relies on two primary categories of data:

- Space-based astronomical datasets obtained through NASA open APIs
- Ground-based optical observations captured using a telescope located in Mauritius

These datasets differ significantly in scale, precision, cadence, and noise characteristics, making their integration a meaningful data engineering challenge.

Data Science Perspective

From a data science standpoint, the project addresses:

- Time-series data analysis
- Data cleaning and normalisation across heterogeneous sources
- Signal comparison and consistency analysis
- Metadata-driven filtering and aggregation
- Exploratory data analysis through visual interfaces

The emphasis is on data structure, behaviour, and interpretation rather than astrophysical discovery.

System Architecture Perspective

The platform is designed as a modular system consisting of:

- A high-performance backend responsible for data ingestion, processing, and storage
- A web-based frontend for authenticated data access and visualisation
- External data providers accessed via APIs
- Local data acquisition from telescope observations

This separation allows independent scaling, testing, and evolution of each component.

Backend Architecture

The backend is designed with performance, predictability, and low-level control as primary concerns and is implemented using C++23.

Key design considerations include:

- Deterministic memory management through explicit ownership models and RAI
- Efficient handling of large astronomical time-series datasets using contiguous data structures
- Low-latency HTTP APIs with minimal runtime overhead
- Strong typing and compile-time guarantees using modern C++ templates and concepts
- Explicit control over concurrency, threading, and synchronization primitives

The backend design considers modern C++23 features such as:

- Concepts and constraints to enforce API correctness at compile time
- `constexpr` and compile-time evaluation to reduce runtime cost
- Structured bindings and ranges for expressive yet efficient data processing

Frontend Architecture

The frontend is designed to be lightweight, reactive, and maintainable, using a server-driven architecture.

Key design aspects include:

- Server-rendered HTML templates as the primary rendering mechanism
- Progressive enhancement using HTMX for partial page updates and dynamic content loading
- Alpine.js for fine-grained client-side state management and UI reactivity
- Minimal JavaScript footprint to reduce complexity and improve performance
- HTTP-based interactions aligned with RESTful backend APIs
- Strategic use of browser caching and conditional requests to minimise redundant data transfers
- Secure authentication flows integrated with server-side session management
- Clear separation between presentation, interaction logic, and data access

Authentication and Access Control

The platform includes user authentication to support:

- Controlled access to datasets
- User-specific observation sessions
- Role-based access for future extensions
- Secure handling of locally captured data

This mirrors real-world scientific platforms where data access is not always fully public.

Mauritius Context

Mauritius provides a unique context for this project due to:

- Its geographic location in the southern hemisphere
- Limited access to large observatories
- Growing academic interest in data-driven research
- Accessibility of amateur-grade optical equipment

The project demonstrates how local resources can contribute meaningfully to global scientific data ecosystems.

Academic Relevance

This project aligns with undergraduate data science objectives by combining:

- Data acquisition and integration
- System and software architecture
- Applied data analysis
- Real-world constraints and trade-offs

It demonstrates that data science extends beyond machine learning into systems, infrastructure, and data-intensive applications.

Expected Outcomes

The expected outcomes of the project include:

- A functional prototype of the observation platform
- Successful ingestion of NASA space data
- Integration of locally captured telescope observations
- An interactive frontend for data exploration
- A documented system architecture suitable for future extension

Conclusion

Nyx aims to demonstrate how modern data platforms can bridge global open data and local data collection. By focusing on architecture, data pipelines, and system design, the project highlights the interdisciplinary nature of data science and its applicability to scientific domains beyond traditional analytics.